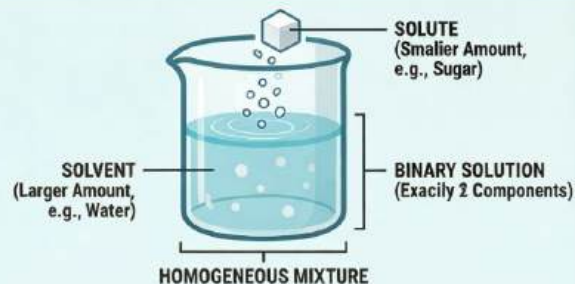


A Complete Guide to Chemical Solutions

THE BASICS OF A SOLUTION

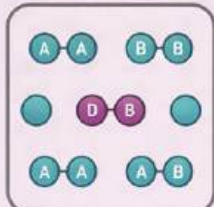


SOLUTE
Component present in a smaller amount in the solution (e.g., sugar)

SOLVENT
Component present in a larger amount in the solution (e.g., water)

IDEAL vs. NON-IDEAL SOLUTIONS

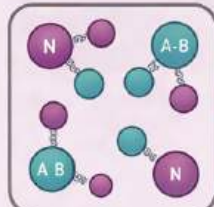
IDEAL SOLUTIONS



✓ Obey Raoult's Law.
 $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$.

POSITIVE DEVIATION
Vapor pressure is higher than predicted. A-B interactions are weaker.
 $\Delta H_{mix} > 0$ (Endothermic),
 $\Delta V_{mix} > 0$.

NON-IDEAL SOLUTIONS



✗ Do not obey Raoult's Law.
Interactions are unequal.
 $\Delta H_{mix} \neq 0$ and $\Delta V_{mix} \neq 0$.

NEGATIVE DEVIATION
Vapor pressure is lower than predicted. A-B interactions are stronger.
 $\Delta H_{mix} < 0$ (Exothermic),
 $\Delta V_{mix} < 0$.

MEASURING CONCENTRATION



MASS PERCENTAGE (%)
$$\left(\frac{\text{Mass of Solute}}{\text{Mass of Solution}} \right) \times 100$$



PARTS PER MILLION (PPM)
$$\left(\frac{\text{Mass of Component}}{\text{Mass of Solution}} \right) \times 10^6$$



MOLARITY (M)
$$\frac{\text{Moles of Solute}}{\text{Volume of Solution (in Liters)}}$$

Note: Molarity is dependent on temperature.



MOLALITY (m)
$$\frac{\text{Moles of Solute}}{\text{Mass of Solvent (in kg)}}$$

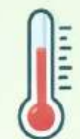
Note: Molality is independent of temperature.

FUNDAMENTAL LAWS & CONCEPTS



MOLE FRACTION (X)
$$\frac{\text{Moles of a Component}}{\text{Total Moles in Solution}}$$

For a binary solution,
 $X_A + X_B = 1$.



EFFECT OF TEMPERATURE ON SOLUBILITY:
• Endothermic processes: Solubility increases with temperature.
• Exothermic processes: Solubility decreases with temperature.



HENRY'S LAW
The partial pressure of a gas in the vapor phase (P) is directly proportional to its mole fraction (X) in the solution.
 $P = K_H \times X$



SOLUBILITY
The maximum amount of solute that can be dissolved in a specific amount of solvent at a particular temperature.

CORRECTING FOR REALITY: ABNORMAL MOLAR MASS

When the molar mass calculated from colligative properties is either lower or higher than the expected normal values. This occurs when solutes associate or dissociate.



VAN'T HOFF FACTOR (i)
A correction factor applied to colligative property formulas to account for the association or dissociation of solute particles.



CALCULATING 'i' FOR DISSOCIATION
$$i = 1 + (n - 1)\alpha$$

where 'n' is the number of particles formed and 'α' is the degree of dissociation.



CALCULATING 'i' FOR ASSOCIATION
$$i = 1 + \left(\frac{1}{n} - 1 \right) \alpha$$

where 'n' is the number of particles associating and 'α' is the degree of association.



COLLIGATIVE PROPERTIES
Properties of a solution that depend only on the number of solute particles, not on their nature or identity.



RELATIVE LOWERING IN VAPOUR PRESSURE

$$\frac{P^\circ - P_1}{P^\circ} = X_2$$

Where P° is the vapor pressure of the pure solvent.



ELEVATION IN BOILING POINT
$$\Delta T_b = K_b \cdot m$$

The boiling point of a solution with a non-volatile solute is higher than the pure solvent.



DEPRESSION IN FREEZING POINT

$$\Delta T_f = K_f \cdot m$$

The freezing point of a solution with a non-volatile solute is lower than the pure solvent.



OSMOTIC PRESSURE (π)
$$\pi = CRT$$

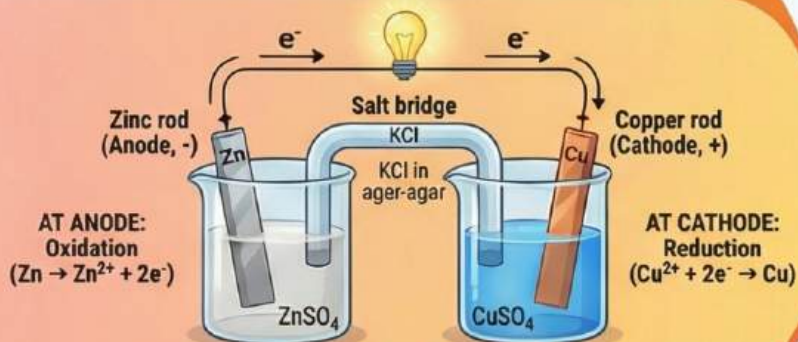
This is the pressure required to stop the process of osmosis (flow of solvent from low to high concentration across a semipermeable membrane).

ELECTROCHEMISTRY UNPLUGGED: A VISUAL GUIDE

@ThunderStudy

Made By Dr Pragya

ELECTROCHEMICAL CELL (VOLTAIC/GALVANIC)



Cell Representation: $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq}) || \text{Cu}^{2+}(\text{aq}) | \text{Cu(s)}$
Current flows from the Copper cathode to the Zinc anode, opposite to the electron flow.

ELECTROCHEMISTRY:
The bridge between chemical and electrical energy.
Involves Chemical ↔ Electrical energy conversions.

ELECTROLYTIC CELL



TWO SIDES OF THE SAME COIN

	Electrochemical Cell	Electrolytic Cell
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Anode Charge	Negative (-)	Positive (+)
Cathode Charge	Positive (+)	Negative (-)
Process	Spontaneous	Non-spontaneous

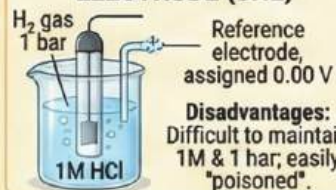
DRIVING FORCE: ELECTRODE & CELL POTENTIAL

Electrode Potential (E)
Potential difference at electrode-electrolyte interface due to charge separation.

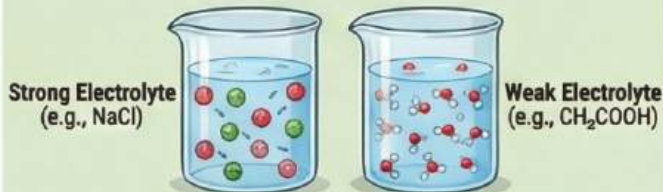
Standard Electrode Potential (E°)
Measured at 1M concentration under standard conditions.

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{(right)}} - E^{\circ}_{\text{(left)}} \\ = E^{\circ}_{\text{(cathode)}} - E^{\circ}_{\text{(anode)}}$$

THE BENCHMARK: STANDARD HYDROGEN ELECTRODE (SHE)



HOW SOLUTIONS CONDUCT ELECTRICITY

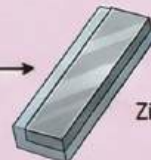


EFFECT OF DILUTION: Conductivity (κ) decreases (fewer ions/volume), Molar Conductivity (Λ_m) increases (total volume increases). For weak electrolytes, Λ_m rises sharply due to increased dissociation.

THE UNWANTED REACTION: CORROSION



Corrosion: Gradual destruction of metals by chemical reactions with environment (e.g., rusting).



A SMART PREVENTION: GALVANIZATION

Iron coated with a more reactive metal (Zinc). Zinc corrodes first. "sacrificing" itself to protect iron (Sacrificial method).

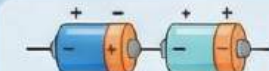
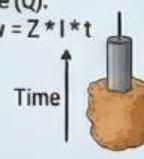
BEYOND THE STANDARD: NERNST EQUATION & GIBBS ENERGY

Nernst Equation (Non-standard conditions):
 $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.0591/n) \cdot \log(Q_c)$
(where n = moles of electrons and Q_c is the reaction quotient: $[\text{products}]/[\text{reactants}]$)

Gibbs Free Energy (ΔG): Relates cell potential to spontaneity.
Standard: $\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}$; Non-Standard: $\Delta G = -nFE_{\text{cell}}$. Negative ΔG indicates spontaneity.

THE LAWS OF ELECTROLYSIS (FARADAY'S LAWS)

Faraday's First Law
Mass (w) of substance deposited w Total electric charge (Q).
Formula: $w = Z \cdot I \cdot t$



Faraday's Second Law
For same charge through different electrolytes in series, mass deposited w chemical equivalent weights (E).
Formula: $\frac{w_1}{E_1} = \frac{w_2}{E_2}$ Understudy

A Crash Course in Chemical Kinetics



THE BASICS OF REACTION RATE

What is Reaction Rate?

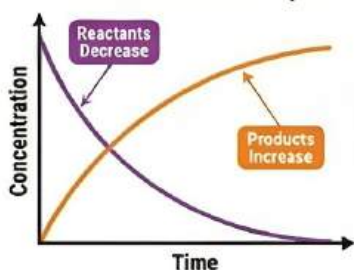
It measures the speed of a reaction, defined as the change in the concentration of a reactant or product over a specific period of time.

General Formula for Rate of Reaction:

$$\text{Rate} = \frac{\text{Change in Concentration}}{\text{Change in Time}}$$

A larger change in concentration over a short time means a faster reaction.

Concentration vs. Time Graphs



Reactants Decrease, Products Increase:

The concentration of reactants decreases over time, while the concentration of products increases. This is a fundamental concept for understanding reaction progress.

EXPRESSING THE RATE OF REACTION

General Chemical Equation



Rate of Disappearance (Reactants)

For A:

$$\text{Rate} = -\frac{1}{3} \cdot \frac{d[A]}{dt}$$

Negative sign indicates a decrease.

Rate of Appearance (Products)

For C:

$$\text{Rate} = +\frac{1}{5} \cdot \frac{d[C]}{dt}$$

Positive sign indicates an increase.

All Rate Expressions are Equal:

For the reaction above,

$$-(1/3) \frac{d[A]}{dt} = -(11) \frac{d[B]}{dt} = +(1/5) \frac{d[C]}{dt}$$

This equality is crucial for solving numerical problems.



Average vs. Instantaneous Rate

Average Rate uses Δ (D) for a time interval $\times$ time interval.



Instantaneous Rate uses the derivative (R) for a specific moment in time.

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RATE LAW & REACTION ORDER

The Rate Law Expression:

An equation that shows how instantaneous rate depends on the concentration of reactions only.

Order "n" is Experimental:

The order of a reaction (n) must be determined through experiments. It is NOT the stoichiometric coefficient.

$$\text{General Rate Law: Rate} = k \cdot [\text{Reactant}]^n$$

Rate Laws in Exponential:

The order of a reaction (n) must be determined through experiments, NOT the stoichiometric coefficient.

Properties of Reaction Order (n):

The order can be a zero, integer, or fraction. For multi-reaction expressions, the overall order is the sum of the individual powers.

$$\text{Unit of Rate Constant (k): } (\text{mol})^{(1-n)} \cdot (\text{L})^{(n-1)} \cdot \text{s}^{-1}$$

FACTORS AFFECTING REACTION RATE



Temperature's Strong Influence

Increasing temperature significantly increases the reaction rate. Rate of thumb: rate approximately doubles for every 10°C rise.

The Arrhenius Equation:

$$k = A \cdot e^{-E_a/RT}$$

Relates rate constant (A) to temperature (T) and activation energy (E_a).



How Catalysts Work

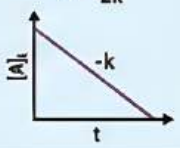
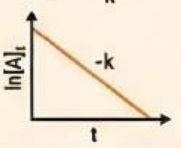
A catalyst increases the reaction rate by providing an alternative reaction pathway with a lower activation energy (E_a).



Collision Theory

For a reaction to occur, reactant molecules must collide with both sufficient energy (equal to or greater than activation energy) and the correct orientation.

INTEGRATED RATE LAWS: ZERO VS. FIRST ORDER

Zero-Order Reaction	First-Order Reaction
Rate Law: $\text{Rate} = k$	Rate Law: $\text{Rate} = k[A]$
Integrated Rate Equation: $[A]_t = -kt + [A]_0$	Integrated Rate Equation: $\ln[A]_t = -kt + \ln[A]_0$
Half-Life ($t_{1/2}$): $t_{1/2} = \frac{[A]_0}{2k}$	Half-Life ($t_{1/2}$): $t_{1/2} = \frac{0.693}{k}$
	
-	Useful Numerical Formula: $t = \frac{2.303}{k} \cdot \log \left(\frac{[A]_0}{[A]_t} \right)$

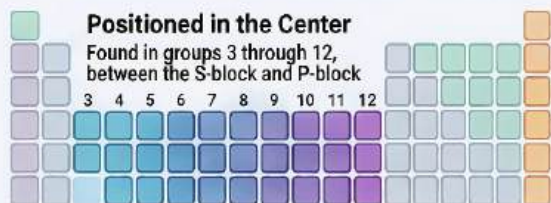
MOLECULARITY VS. ORDER: A KEY DISTINCTION

Molecularity	Order of Reaction
	
Definition: Number of reactant species colliding in an elementary step.	Definition: Sum of powers of concentration terms in the rate law.
Determination: Theoretically, based on the reaction mechanism.	Determination: Experimental, must be measured.
Possible Values: Positive integers (1, 2, rarely 3). Cannot be zero or fractional.	Possible Values: Can be zero, positive, negative, or fractional.
Applicability: Defined only for elementary (single-step) reactions.	Applicability: Defined for both elementary and complex reactions.
Molecularity Limit: Molecularity is almost never greater than 3, as the probability of a simultaneous, effective collision between more than three particles is extremely low.	Slowest Step is Key: In a multi-step reaction, the overall rate is determined by the slowest step, known as the rate-determining step.

The Ultimate Guide to D & F Block Elements

@ThunderStudy

THE WORLD OF D-BLOCK ELEMENTS



Why "Transition" Elements?

Form a bridge between highly electropositive S block metals and less electropositive P-block elements.



The Official Definition

An element with an incompletely filled d-orbital in its atomic state or any common oxidation state.



The Exception: Zinc (Zn), Cadmium (Cd) & Mercury (Hg)

Not considered transition elements; have completely filled d-orbital (d^{10}) in both atomic and common +2 oxidation states.

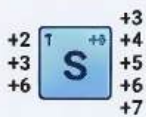
KEY TRENDS AND PROPERTIES

Atomic Radii Trend Across the Series



Decreases (increased nuclear charge), then almost constant (electron repulsion balances charge), finally increases slightly (repulsion dominates).

Variable Oxidation States



Caused by small energy difference between nd and $(n-1)d$ orbitals, allowing electrons from both to participate in bonding (Mn shows +2 to +7).

Why are They Great Catalysts?



Variable oxidation states allow participation in diverse chemical reactions (e.g., Iron, V_2O_5).

Ionization Enthalpy: The Cost of Removing an Electron



Energy required to remove an electron. High for stable half-filled (d^5) or fully-filled (d^{10}) configurations.

THE SUPER TRICK: IT'S ALL ABOUT UNPAIRED ELECTRONS

The More Unpaired Electrons, the Stronger the Properties

Unpaired d-orbital electrons are key to many properties.

Chain of Effects from Unpaired Electrons



Strong Metallic Bonding



Hardness:
Metals are very hard and tough.



High Melting Points:
Requires a lot of energy to break bonds.



High Enthalpy of Atomization:
Difficult to break into individual atoms.



Paramagnetism:
Attracted by a magnetic field.



Colored Compounds:
Variety of colors due to d-d electron transitions.



The Opposite: Zero Unpaired Electrons

Full d^{10} configuration leads to weak metallic bonding, softer metals, lower melting points (e.g., Mercury is liquid).



The Manganese (Mn) Anomaly

Despite d^5 (many unpaired electrons), has unusually low melting point: stable, stable, half-filled configuration makes it reluctant to form strong metallic bonds.



A QUICK LOOK AT THE F-BLOCK

Two Series: Lanthanoids and Actinoids

Lanthanoids: 14 elements following Lanthanum (La).
Actinoids: 14 elements following Actinium (Ac).

Lanthanoid Contraction: A Crucial Concept



Cause: Poor shielding effect of 4f-electrons; nucleus pulls outer electrons more strongly, causing the atom to shrink.

Lanthanoid Contraction

Elements of 4d and 5d series have nearly identical atomic radii and properties (e.g., Zirconium and Hafnium are almost chemical twins).

Consequence of Lanthanoid Contraction



Elements of 4d and 5d series have nearly identical atomic radii and properties (e.g., Zr and Hf are almost chemical twins).

Actinoid Contraction

Similar size decrease in Actinoid series due to poor shielding effect of 5f-electrons.

Mutual Donation
(Strengthens Bond)

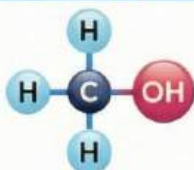
Mutual Donation
(Strengthens Bond)

A Visual Guide to Alcohols, Phenols & Ethers

@ThunderStudy

1. Meet the Functional Groups

Alcohol (R-OH)



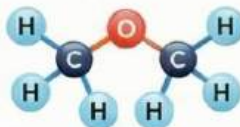
An -OH group is attached to a saturated carbon atom (part of an alkyl group, R).
Example: $\text{CH}_3\text{-OH}$ (Methanol).

Phenol (Ar-OH)



An -OH group is directly attached to a carbon atom within a benzene ring.
Example: Benzene ring with an -OH group.

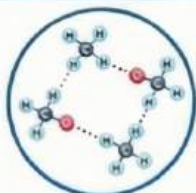
Ether (R-O-R')



An oxygen atom is bonded to two alkyl or aryl groups.
Example: $\text{CH}_3\text{-O-CH}_3$ (Dimethyl ether).

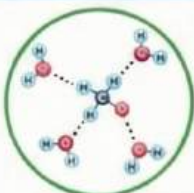
2. The World of Alcohols

Properties



High Boiling Points

Due to intermolecular hydrogen bonding between -OH groups. Boiling point decreases with increased branching.



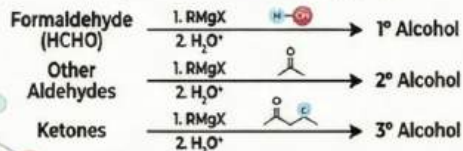
Soluble in Water

Smaller alcohols form hydrogen bonds with water molecules.

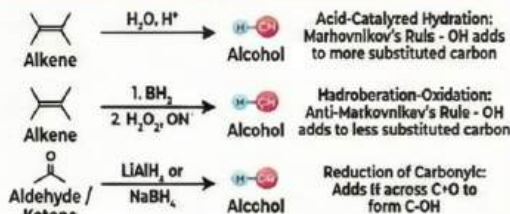
Acidity

Phenol > Water > Alcohol

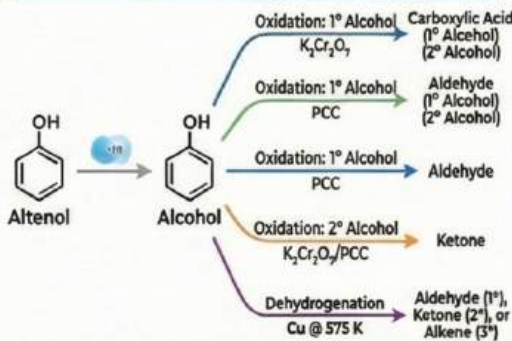
Alcohols are less acidic than phenols and water, partly due to the +I effect of alkyl groups.



Key Preparation Methods

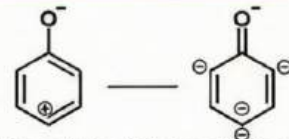


Key Reactions of Alcohols



3. The Special Case of Phenols

Significantly More Acidic than Alcohols



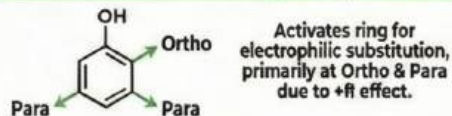
Phenoxide ion is highly stable due to resonance delocalization.

EWG (e.g., $-\text{NO}_2$) Increase Acidity
Pull electron density, stabilize phenoxide.

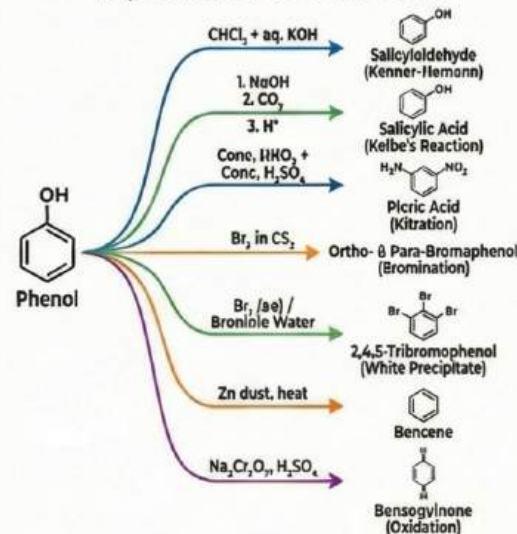
Order: Para > Ortho > Meta.

EDG (e.g., $-\text{CH}_3$) Decrease Acidity
Push electron density, destabilize phenoxide.

Directs Incoming Groups

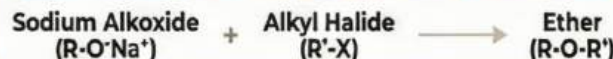


Important Reactions of Phenol



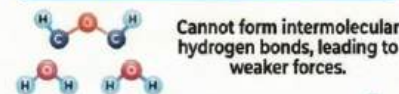
4. Understanding Ethers

Premier Preparation: Williamson Synthesis



Rule: Big Alkoxide, Small Halide for best results.

Lower Boiling Points



5. How to Tell Them Apart: Chemical Tests

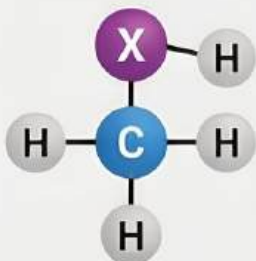
Test 1: Neutral FeCl_3 Test



Test 2: Bromine Water Test



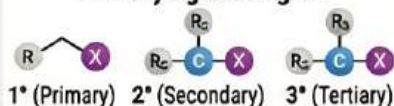
A Visual Guide to Haloalkanes & Haloarenes



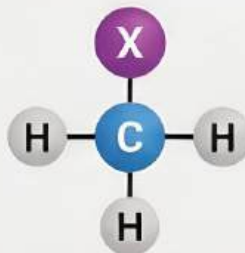
Haloalkane

Alkane with one or more hydrogens replaced by a halogen (F, Cl, Br, I).
General formula: $R-X$.

Identifying the Degree



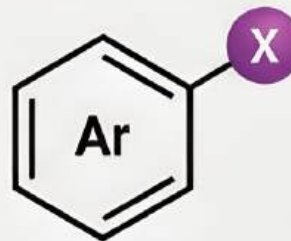
The Basics: Identifying the Compounds



Haloarene

Aromatic compound (like benzene) where a halogen is directly attached.
General formula: $Ar-X$.

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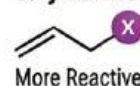


Know Your Halides: Vinyl vs. Allyl

Vinyl Halide



Allyl Halide

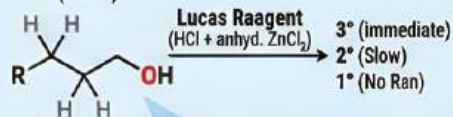


Benzyl Halide

Highly Reactive

How to Make Haloalkanes (Preparation Methods)

From Alcohols ($R-OH$)



The Best Method:
Thionyl Chloride ($SOCl_2$)
→ Preferred
(SO_2 & HCl are gases = Pure product)

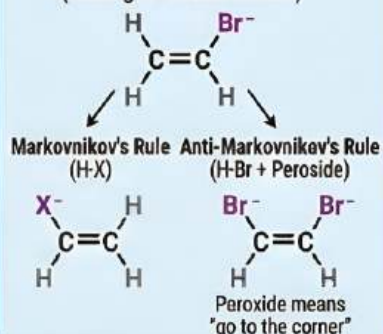
PCl_2

PCl_2

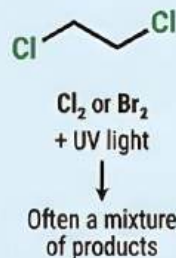
Red P/X_2

$NaBr + H_2SO_4$

From Alkenes (Adding to Double Bonds)



From Alkanes (Free-Radical Halogenation)



Key Mechanisms & Named Reactions

S_N1 vs. S_N2 Reaction Mechanisms

Steps: One Step

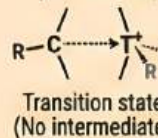
Intermediate: Carbocation formed $-C^+$

Reactivity Order: $3^\circ > 2^\circ > 1^\circ$ (Stable C^+)

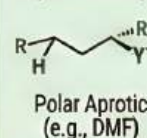
Deciding Factor: Steric Hindrance (Crowding)

Solvent: Polar Protic (e.g., H_2O)

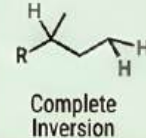
S_N1 (Unimolecular)



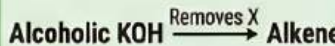
S_N2 (Bimolecular)



Stereochemistry:

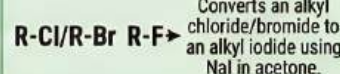


Elimination Reaction (Dehydrohalogenation)



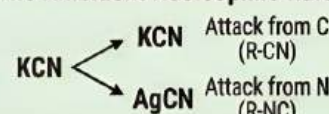
(Mnemonic: Finkelstein uses NaI in Acetone)

Finkelstein Reaction (Halogen Exchange)



(Mnemonic: Swarts = Swag = AgF (Ag Fire))

The Ambident Nucleophile Rule



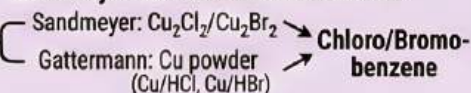
(Mnemonic: When Ag is present, Nitrogen attacks)

Haloarenes: Preparation & Reactivity

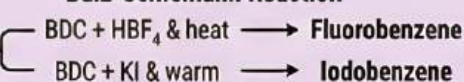
Making Haloarenes from BDC ("The Hero")



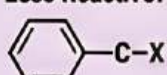
Sandmeyer vs. Gattermann Reaction



Balz-Schiemann Reaction

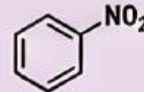


Why are Haloarenes Less Reactive?



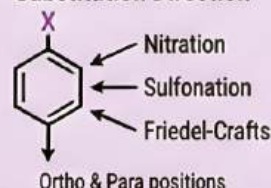
→ Resonance gives $C-X$ bond partial double-bond character (Stronger, harder to break)

Increasing Reactivity

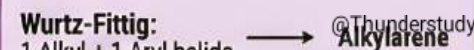
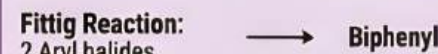
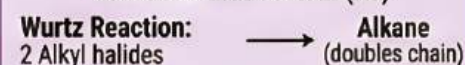


$-NO_2$ at ortho/para positions
→ Makes ring more reactive towards nucleophilic substitution

Electrophilic Substitution Direction



Reactions with Sodium (Na)

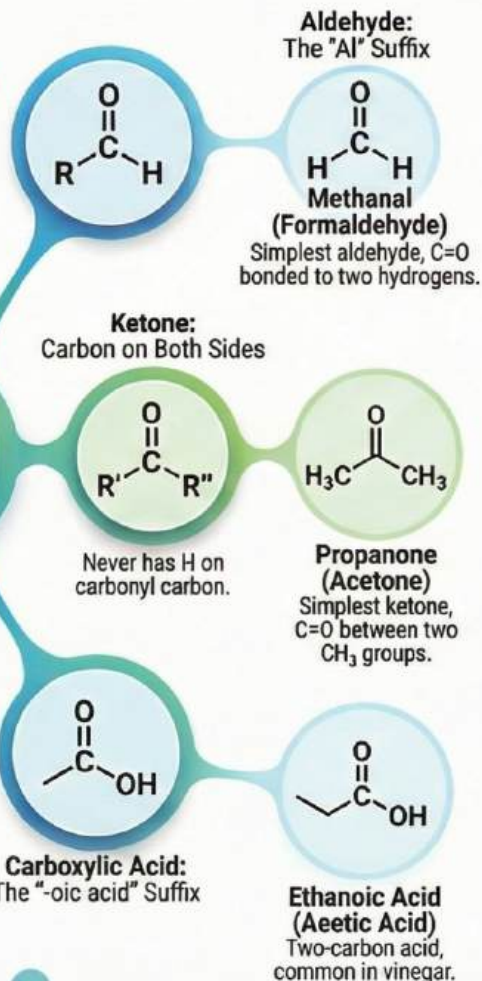


Mastering Aldehydes & Ketones

Part 1: Structure & Preparation

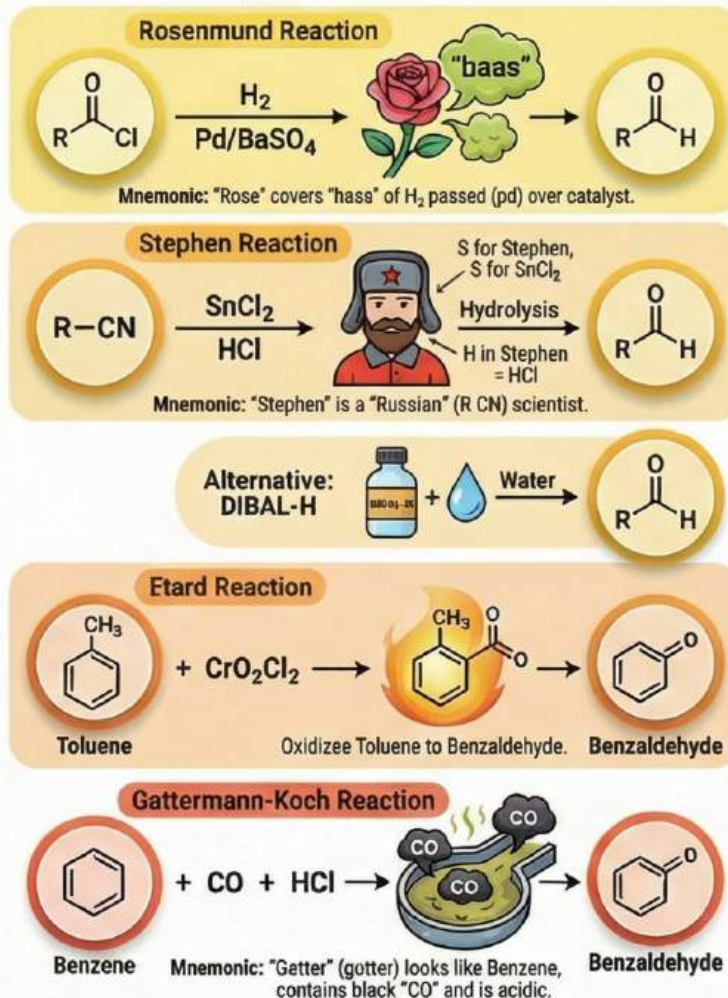
Meet the Carbonyl Family:

Identifying the Key Players



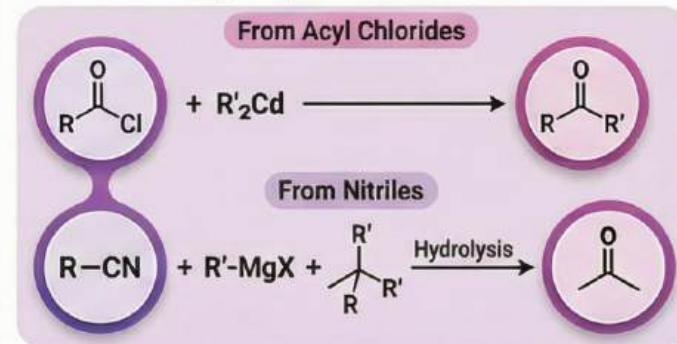
How to Make Aldehydes:

Key Preparation Methods



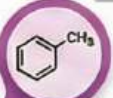
How to Make Ketones:

Key Preparation Methods



Master the Benzene Derivatives:

A Naming Guide

Structure Name	Formula	Description
 Toluene	$\text{C}_6\text{H}_5-\text{CH}_3$	Benzene with a methyl group.
 $\text{C}_6\text{H}_5-\text{CH}_2-$	$\text{C}_6\text{H}_5-\text{CH}_2-$	One H removed from methyl group.
 $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$	$\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$	Two H removed from methyl group.
 Benzo	$\text{C}_6\text{H}_5-\text{C}\equiv$	All three H removed from methyl group. @Thunderstudy

A Visual Guide to Aldehydes, Ketones & Carboxylic Acids

IDENTIFYING THE KEY PLAYERS



ALDEHYDE R-CHO

Characterized by a carbonyl group (C=O) bonded to at least one hydrogen atom.

KETONE -one

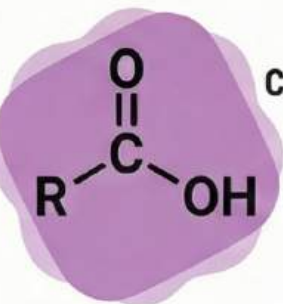
Features a carbonyl group (C=O) bonded to two carbon atoms.



@ThunderStudy

CARBOXYLIC ACID -oic acid

Contains a carbonyl group (C=O) attached to a hydroxyl group (-OH).



HOW TO PREPARE ALDEHYDES & KETONES

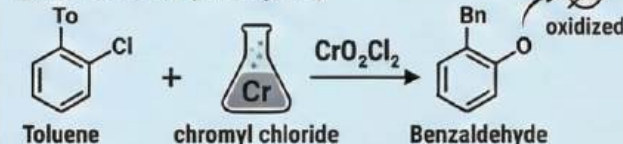
Rosenmund Reaction (Aldehydes)



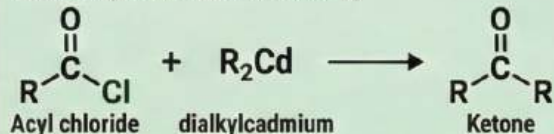
Stephen Reaction (Aldehydes)



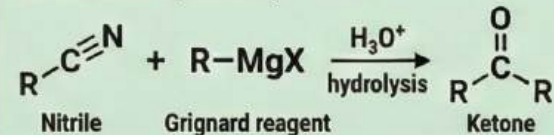
Etard Reaction (Aldehydes)



From Acyl Chlorides (Ketones)

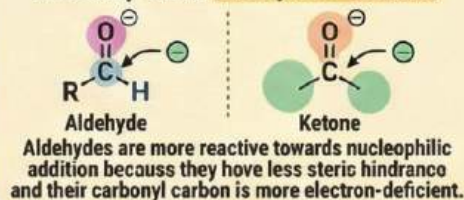


From Nitriles (Ketones)

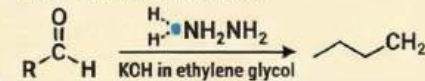


KEY REACTIONS OF ALDEHYDES & KETONES

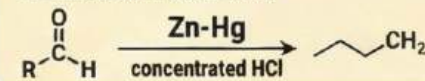
Reactivity Order: Aldehydes > Ketones



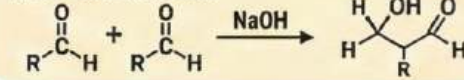
Wolf-Kishner Reduction



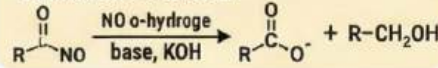
Clemmensen Reduction



Aldol Condensation



Cannizzaro Reaction



KEY DISTINGUISHING TESTS FOR ALDEHYDES

'Tollens' Test

Reagent: Silver Nitrate

Ammoniacal Silver Nitrate



Positive Result for Aldehydes: Silver Mirror

Works on Aromatic Aldehydes? Yes

Fehling's Test

Reagent: Solution A + B

Fehling's Solution A + B



Red-brown Precipitate (Cu₂O)

No

Iodoform Test

Reagent: I₂ / NaOH

Yellow Precipitate

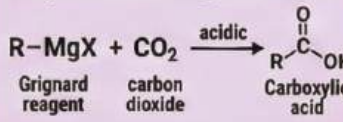


Yellow Precipitate (CHI₃)

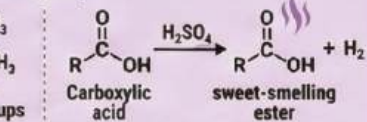
Only for compounds with a CH₃-CO group

ALL ABOUT CARBOXYLIC ACIDS

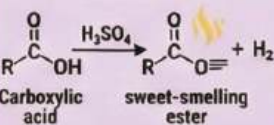
Preparation from Grignard Reagent



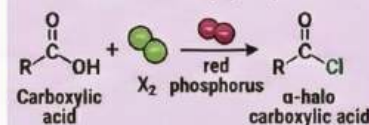
Acidity depends on attached groups



Esterification



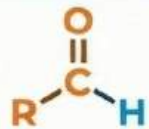
Hell-Volhard-Zelinsky (HVZ) Reaction



Distinguishing from Phenol

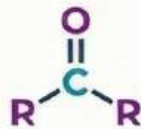


A Visual Guide to Aldehydes, Ketones & Carboxylic Acids

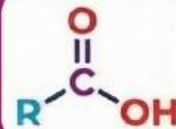


ALDEHYDE:
The "-al" Family
Carbonyl (C=O) bonded to at least one H. Suffix "-al".
e.g., Methanal, Ethanal.

IDENTIFYING THE KEY PLAYERS



KETONE:
The "-one" Family
Carbonyl (C=O) bonded to two carbon atoms. Suffix "-one".
e.g., Propanone/Acetone.

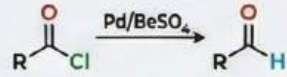


CARBOXYLIC ACID:
The "-oic acid" Family
Carboxyl (C=O) attached to hydroxyl (-OH). Suffix "-oic acid".
e.g., Ethanoic Acid.

PREPARATION OF ALDEHYDES

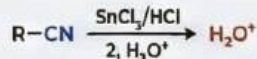


Resenmund Reaction



Acyl Chloride hydrogenated using H_2 with Pd/BaSO₄ catalyst.

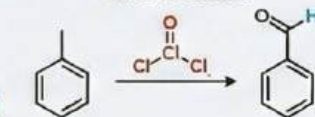
Stephen Reaction



Nitriles reduced with SnCl₄/HCl then hydrolyzed. Mnemonic: "Stephen is a Russian (R CN) Scientist".

Alternative to Stephen Reaction
DIBAL H followed by hydrolysis can also convert nitriles to aldehydes.

Etard Reaction



Toluene is oxidized by chromyl chloride (CrO₂Cl₂) to Benzaldehyde.

Cattermann-Koch Reaction



Benzene treated with CO, HCl in presence of AlCl₃.
Mnemonic: A drain cuser (Catter) looks like Benzene, containing black CO gas and acidic HCl.

PREPARATION OF KETONES



From Acyl Chlorides
Two moles of Acyl Chloride react with Dialkyl Cadmium (R₂Cd) to form a ketone.



From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O*) to yield a ketone.



From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O*) to yield a ketone.

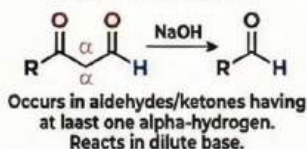


From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O*) to yield a ketone.

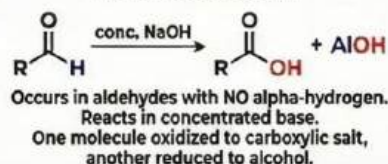
KEY REACTIONS & TESTS

	Tollens' Test	Fehling's Test	Iodoform Test
Test	Ammoniacal Silver Nitrate [Ag(NH ₃) ₂] ⁺ OH ⁻	Fehling's Solution (Cu ²⁺ ions)	I ₂ and NaOH
Outcome	Aldehydes Only Silver Mirror	Aliphatic Aldehydes Only Red-Brown Precipitate	Compounds with CH ₃ -C=O group Yellow Precipitate
Fact	Both aliphatic & aromatic aldehydes give this test.	Aromatic aldehydes (like Benzaldehyde) do NOT give this test.	Used to detect methyl ketones and acetaldehyde.

Aldol Condensation

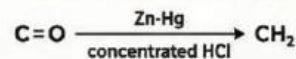


Cannizzaro Reaction



Wolff-Kishner Reduction using Hydrazine (NH₂-NH₂) and a strong base (KOH)

Clemmensen Reduction



ALL ABOUT CARBOXYLIC ACIDS

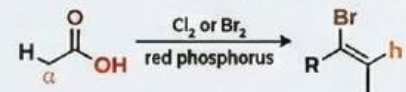
Comparing Acidic Strength

↑ Acidity increases with electron-withdrawing groups (-I effect, e.g., -NO₂, -F, -Cl)
↓ Acidity decreases with electron-donating groups (+I effect, e.g., -CH₃)

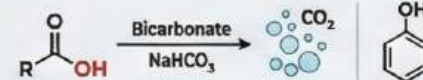
Relationship with pK_a

Carboxylic acids with an alpha-hydrogen react with Cl₂ or Br₂ in the presence of red phosphorus to substitute a halogen on the alpha-carbon.

Hell-Volhard-Zelinsky (HVZ) Reaction



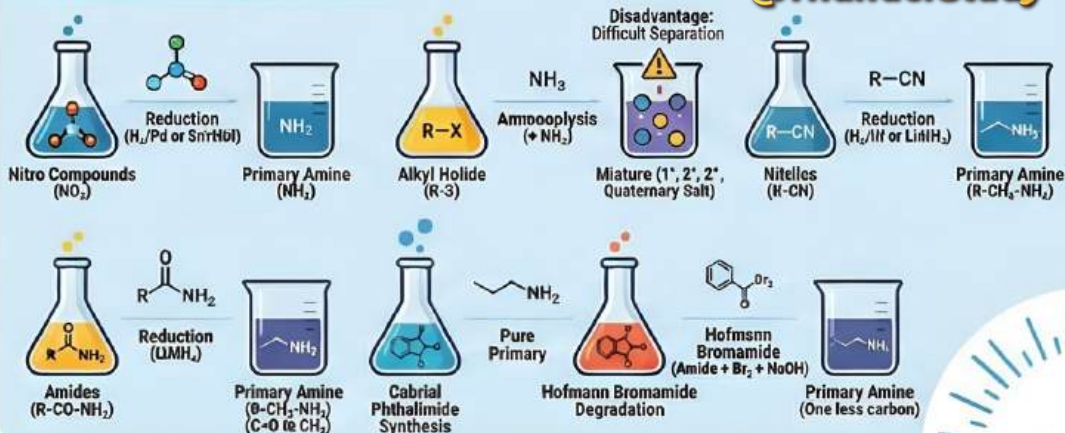
Test to Distinguish from Phenol



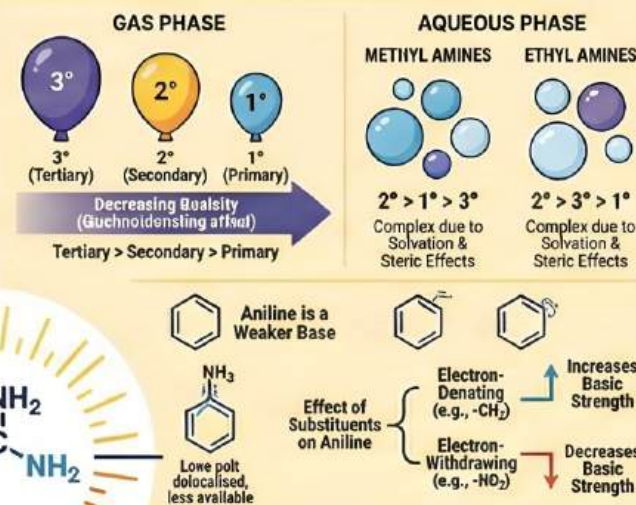
Carboxylic acids react with Sodium Bicarbonate (NaHCO₃) to produce brisk effervescence (CO₂ gas). Phenols do not.

Mastering Amines: A Visual Guide to Key Reactions & Concepts

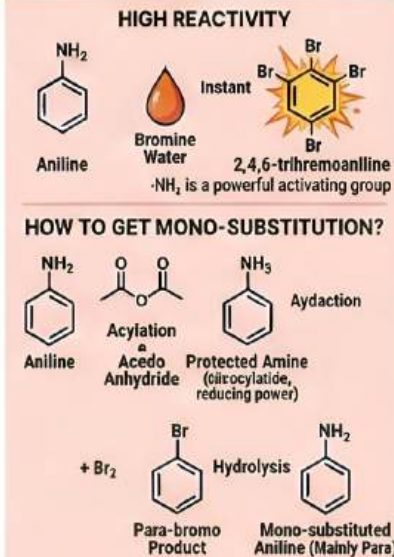
1. PREPARATION OF AMINES



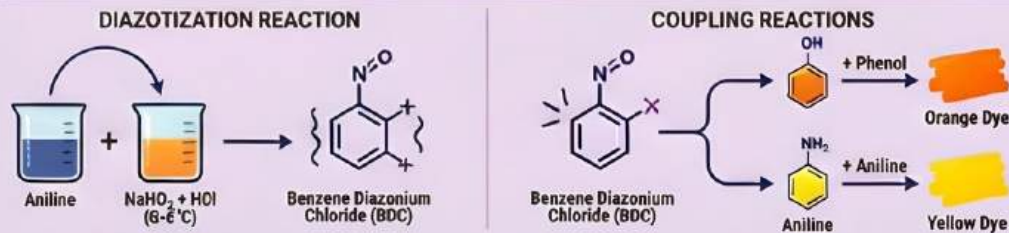
2. UNDERSTANDING BASIC STRENGTH



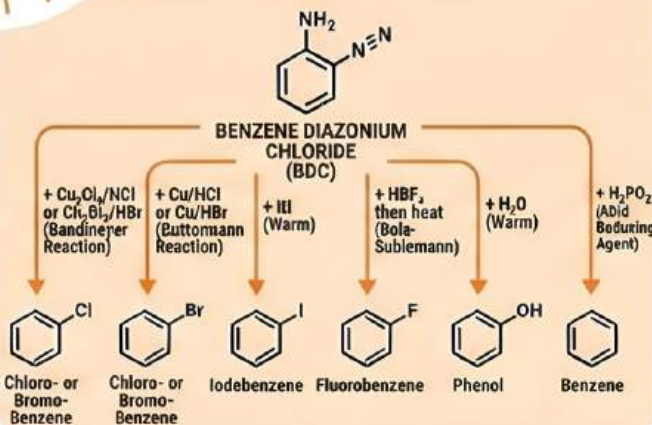
5. ELECTROPHILIC SUBSTITUTION OF ANILINE



3. KEY CHEMICAL REACTIONS & TESTS



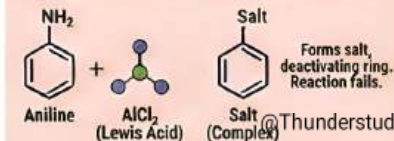
4. THE VERSATILITY OF BDC



THE ANOMALY OF NITRATION



WHY ANILINE FAILS FRIEDEL-CRAFTS



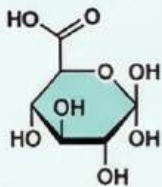
Biomolecules Unpacked: A Visual Guide to Carbohydrates & Proteins

CARBOHYDRATES

@ThunderStudy

PROTEINS

1. What are Carbohydrates?



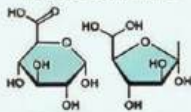
The 3 Core Conditions

The Chemical Definition
Optically active polyhydroxy aldehydes or ketones.



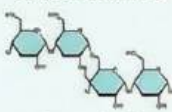
2. How Carbohydrates are Classified

Monosaccharides



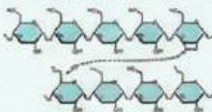
Glucose - Fructose
Simplest units, cannot be hydrolyzed. Single units.

Oligosaccharides



Oligosaccharides
Hold 2 to 10 monosaccharide units on hydrolysis.

Polysaccharides



Starch, Cellulose
Yield a large number of monosaccharide units. Complex polymers.

3. Reducing vs. Non-Reducing Sugars

Reducing Sugars



Have free aldehyde/ketone group, Reduce Tollens' or Fehling's reagents. Examples: Glucose, Fructose.

Non-Reducing Sugars



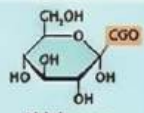
No free aldehyde/ketone group. Example: Sucrose.

4. A Closer Look: Glucose & Fructose

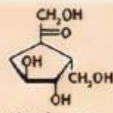


Identical Formula

The Key Structural Difference



Aldohexose



Ketohexose



Glucose: "Grape sugar"



Fructose: Fruits, Honey

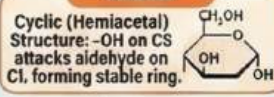
6. The Failure of the Linear Structure & Rise of the Cyclic Form

Problem

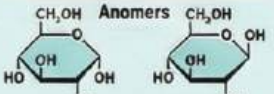


These two cyclic forms, which differ only in the configuration of the -OH group on the first carbon (C1), are called anomers.

Solution



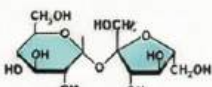
Cyclic (Hemiacetal) Structure: -OH on C1 attacks aldehyde on C1, forming stable ring.



Alpha (α)

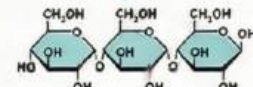
Beta (β)

7. Building Larger Carbohydrates



Disaccharide: Sucrose
 α -D-Glucose + β -D-Fructose

Glycosidic Linkage:
Oxide linkage between monosaccharides with water removal.



Polysaccharide: Starch & Cellulose
Starch: polymer of α glucose, storage in plants
Cellulose: straight chain of β -glucose, plant cell walls

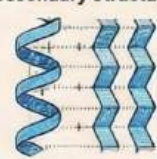
9. The Four Levels of Protein Structure

Primary Structure



Specific sequence of amino acids in polypeptide chain.

Secondary Structure



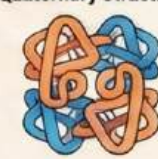
Local shape: α -Helix or β -Pleated Sheet, stabilized by hydrogen bonds.

Tertiary Structure



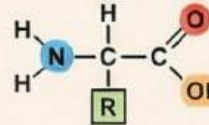
Overall 3D folding of polypeptide chain.

Quaternary Structure

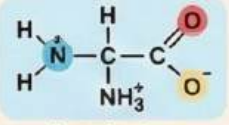


Arrangement of multiple polypeptide chains (subunits).

8. Amino Acids: The Building Blocks



Molecules containing both basic amino and acidic carboxyl groups.



The Zwitterion
In aqueous solution, forms dipolar ion. High melting points.

Essential vs. Non-Essential



Essential

e.g., Valine
Cannot be synthesized by body. Must be obtained from diet.



Non-Essential

e.g., Glycine
Can be made by the body.

10. Denaturation: When Proteins Unravel



Before

Process where protein loses biological activity due to disruption of secondary & tertiary structures.



After



Caused by heat or pH changes. Primary structure remains intact.



Real-World Example:
Coagulation of egg white when boiled.